



Consensus on the Rational Use of Antithrombotics in Veterinary Critical Care (CURATIVE): Domain 3—Defining antithrombotic protocols

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Abstract

Objectives: To systematically examine the evidence for use of a specific protocol (dose, frequency, route) of selected antithrombotic drugs, in comparisons to no therapy or to other antithrombotic therapies, to reduce the risk of complications or improve outcomes in dogs and cats at risk for thrombosis.

Design: Standardized, systematic evaluation of the literature, categorization of relevant articles according to level of evidence (LOE) and quality (Good, Fair, or Poor), and development of consensus on conclusions via a Delphi-style survey for application of the concepts to clinical practice.

Settings: Academic and referral veterinary medical centers.

Results: Databases searched included Medline via PubMed and CAB abstracts. Eight different antithrombotic drugs were investigated using a standardized Patient, Intervention, Comparison, Outcome (PICO) question format both for dogs and cats, including aspirin, clopidogrel, warfarin, unfractionated heparin (UFH), dalteparin, enoxaparin, fondaparinux, and rivaroxaban, generating a total of 16 worksheets. Most studies identified were experimental controlled laboratory studies in companion animals (LOE 3) with only four randomized controlled clinical trials in companion animals (LOE 1).

Conclusions: Overall, evidence-based recommendations concerning specific protocols could not be formulated for most antithrombotic drugs evaluated, either because of the wide range of dosage reported (eg, aspirin in dogs) or the lack of evidence in the current literature. However, clopidogrel administration in dogs and cats at risk of arterial thrombosis, notably in cats at risk of cardiogenic thromboembolism, is supported by the literature, and specific protocols were recommended. Comparably, aspirin should not be used as a sole antithrombotic in cats with cardiomyopathy. Using the available safety profile information contained in the literature, the panel reached consensus on suggested dosage schemes for most antithrombotics. Significant knowledge gaps were highlighted, which will hopefully drive novel research.

KEYWORDS

administration and dosage, anticoagulant, antiplatelet agents, CURATIVE, veterinary

Abbreviations: aPTT, activated partial thromboplastin time; ATE, arterial thromboembolism; IMHA, immune-mediated hemolytic anemia; GI, gastrointestinal; LOE, level of evidence; PICO, acronym stands for P-patient, problem or population. I—intervention. C—comparison, control or comparator. O—outcome; PT, prothrombin time; RCT, randomized controlled trial; UFH, unfractionated heparin; VTE, venous thromboembolism

INTRODUCTION

Despite the widespread use of antithrombotic drugs for the prevention and treatment of arterial and venous thrombosis, thromboembolic diseases continue to be a major cause of death and morbidity both in human,^{1,2} and veterinary medicine.^{3–6} Antithrombotic drugs are

prescribed to inhibit the process of thrombus formation both to prevent and to treat pathologic thrombosis, but the risk-benefit profiles of particular drugs in individual patients should be considered prior to antithrombotic drug selection. The optimal dosing regimens for each drug must be determined for each species, and cannot simply be extrapolated from the human literature. In addition, variation in disease process and severity, in drug metabolism and in individual patient responses to treatment must also be considered, and will likely be multifactorial (eg, breed, sex, age, genetics).⁷

Domains 1 and 2 of the CURATIVE guidelines sought to define the populations of dogs and cats at risk for thrombosis, and to establish the most appropriate drug(s) for specific clinical settings. The objective of Domain 3 was to determine whether some antithrombotic protocols are more effective than others in order to make recommendations about drug dose, and the route and frequency of administration. In addition, Domain 3 also sought to highlight potential adverse effects and to evaluate safety profiles in order to provide guidance to clinicians regarding the risks associated with specific drug choices.

To undertake this task, a standardized Patient, Intervention, Comparison, Outcome (PICO) question was formulated to enable a systematic review of the evidence for 8 antithrombotic drugs (aspirin, clopidogrel, warfarin, unfractionated heparin (UFH), dalteparin, enoxaparin, fondaparinux, and rivaroxaban) in dogs and cats at risk of thrombosis. While UFH was considered as a single entity despite the variation in polysaccharide chain lengths inherent to that product, separate individual low molecular-weight heparin (LMWH) products were considered individually because they do not have identical pharmacologic profiles.

The number of articles considered relevant varied considerably between drugs and species. Specifically, the following numbers of articles were considered relevant for each medication: aspirin (dogs 47, cats 14), clopidogrel (dogs 11, cats 8), warfarin (dogs 16, cats 5), UFH (dogs 8, cats 1), dalteparin (dogs 7, cats 6), enoxaparin (dogs 5, cats 3), fondaparinux (dogs 0, cats 1), and rivaroxaban (dogs 3, cats 1). With the exception of aspirin in dogs, where the literature was extensive, worksheet authors included studies assessing the *in vitro* effect on platelet function and on coagulation assays in addition to pharmacokinetic and pharmacodynamic modeling in healthy animals.

The standardized PICO question read as follows:

In dogs and cats at risk of arterial/venous thrombosis (P), does use of a specific protocol (dose, frequency, route) for use of [antithrombotic drug] (I) compared to no therapy or to other antithrombotic therapies (C) reduce the risk of complications (eg, fatal or nonfatal hemorrhage, transfusion requirements) or improve any outcomes (eg, survival, organ dysfunction, thrombosis) (O)?

PICO QUESTION: Aspirin therapy

Guidelines

3.1 Aspirin (Dogs)

- a. We suggest that oral aspirin may be effective for prevention of arterial thromboembolism (ATE) in dogs.

- b. No evidence-based recommendations can be made for a specific aspirin dosage in dogs.
- c. We suggest that aspirin be given for 2–3 days before full therapeutic effects of aspirin are anticipated, although commencement of aspirin therapy after an arterial insult may still be effective at preventing thrombosis.
- d. No recommendations can be made for, or against, use of aspirin for VTE (venous thromboembolism) in dogs.

3.2 Aspirin (Cats)

- a. We recommend against aspirin as a sole antithrombotic in cats at risk for ATE.
- b. No recommendations can be made concerning appropriate aspirin dosage in cats.

Evidence summary

Dogs

A total of 47 articles were considered relevant to the PICO question, including 20 studies suggesting that protocolized aspirin therapy is more effective, or safer than, no aspirin therapy,^{8–29} or other thromboprophylaxis protocols. Twenty-five studies were considered neutral to the question.^{12,15,16,25–27,30–52} The majority of the published literature regarding the administration of aspirin in dogs involves experimental (laboratory) studies (LOE 3) of arterial thrombosis,^{10,18,22–24,26,28,31,45,53} fewer studies evaluated venous thrombosis.^{41,46–48} Most were considered of fair quality given that controls were included.^{13,15,16,18–20,22–27} Only a few studies used randomized controls, and thus were considered good quality evidence.^{8–12} There were also several studies without controls, considered of poor methodological quality.^{17,28}

The majority of studies were designed to model the treatment of human diseases including coronary artery disease and arterial grafting. Models included coronary angioplasty and endarterectomy-induced endothelial injury,^{10,18,22–24,26,28,53,54} as well as evaluations of the effect of thromboprophylaxis on the rate of coronary arterial reocclusion after thrombolysis.^{11,27,30,34,36,37,49,51} Most of these models are not directly applicable to veterinary clinical medicine. In addition, many studies evaluated only short-term outcome measures used IV aspirin formulations that are not commercially available. Within endarterectomy and angioplasty models in dogs, the difference in aspirin efficacy appears mostly related to the timing of dosing relative to the arterial injury. Aspirin therapy, including aspirin at 5 mg/kg/day,¹⁸ was more likely to be efficacious when started >2 days prior to arterial injury.^{18,28}

Studies investigating graft thrombosis are perhaps most relevant to veterinary patients with ongoing risk factors for thromboembolism, since they involve medium to long-term oral aspirin administration. Twelve studies in graft models supported the research question, but the aspirin dose used varied widely. Unfortunately, in some studies, interpretation of aspirin dosing is impossible because dog bodyweights were not reported, and a standard mg/day dose was administered, rather than a mg/kg/day dose.^{8,17} Of most relevance were the



limited number of graft studies that directly compared low-dose (1–5 mg/kg/day) to high-dose (>13 mg/kg/day) aspirin therapy, but again, results varied.^{12,20,26}

For instance, in a study of femoral arterial graft patency (LOE 3, Fair), high-dose (13 mg/kg/day) aspirin but not low-dose aspirin (1 mg/kg/day) improved 1-week graft patency when commenced 24 hours prior to prosthesis implantation and continued for the duration of the experiment.²⁰ In a similar study (LOE 3, Good) McDaniel et al,¹² found that high-dose oral aspirin (13.5–18 mg/kg/day) with dipyridamole (3 mg/kg/day), but not low-dose aspirin (1 mg/kg/day) improved 7-day graft patency compared to control when commenced 24 hours prior to carotid artery grafting.¹² In contrast, Escudero et al,²⁶ (LOE 3, Fair) found that 5 mg/kg/day of oral aspirin was the only effective treatment for reducing platelet deposition on 6 biomaterials used as ex vivo arterial shunts, when commenced 7 days before implantation.²⁶ Specifically, 5 mg/kg/day of aspirin was more effective than 20 mg/kg/day of aspirin, 5 mg/kg/day aspirin combined with 5 mg/kg/day dipyridamole, and negative control. The apparent difference in efficacy of aspirin between studies may be related to the duration of aspirin therapy prior to shunt placement. Nonetheless, the limitations of studies such as that by Escudero et al are that they only evaluate immediate platelet deposition or thrombosis, rather than over a clinically relevant time period.

Thirteen studies involving graft models were considered neutral to the PICO question because they lacked a nonaspirin control group that precluded assessment of the efficacy of aspirin. No clinical adverse effects of aspirin were noted in these studies.^{15,38–40,42–44,46–48,50,55,56} An additional 2 studies were considered neutral because there was no significant difference in the thromboprophylactic efficacy between aspirin treated groups, with doses ranging from 1–50 mg/kg/day.^{14,33}

Only 2 studies evaluated the antithrombotic effects of aspirin in a clinical population of dogs at risk of thrombosis. Mellet et al (LOE 2, Good) performed a single-center unblinded, randomized, positive-controlled clinical trial in which low dose oral aspirin (0.5 mg/kg/day) alone was compared to clopidogrel alone or a combination of clopidogrel and aspirin for thromboprophylaxis in dogs with immune-mediated hemolytic anemia (IMHA).⁵⁷ Twenty-four dogs were randomized to either treatment and no difference was found in survival to discharge or 90 days survival, rate of thrombosis, or other adverse effects. As such the effects of aspirin were considered to be neutral to the PICO question. One patient in each of the aspirin and clopidogrel groups was suspected to have died of a thrombotic complication. Unfortunately, this study was likely underpowered to detect a difference between groups since sample size was based on a survival difference of 50% between groups, which is likely an optimistic target in a complex disease such as IMHA. One study (LOE 4, Poor) supported the PICO question.²⁹ In this single center retrospective study of dogs with IMHA, aspirin (0.5 mg/kg/day) and azathioprine in addition to glucocorticoid therapy resulted in improved survival to hospital discharge and to 30 days compared to dogs that did not receive thromboprophylaxis, and to those that received azathioprine and UFH.²⁹

Several studies demonstrated that a single IV aspirin dose of 10–15 mg/kg shortly after arterial intimal injury resulted in a lower incidence

of occlusive thrombus formation when compared to control.^{22,23,53} Similarly, Hernandez-Maldonado et al,²⁴ (LOE 3, Fair) reported that oral aspirin (10 mg/kg/day) was more effective than control in preventing thrombosis in injured femoral arteries in a model of extremity trauma, when commenced on the first day postoperatively and continued for 3 weeks.

Multiple studies (LOE 3, Fair) assessed the in vitro function of platelets when dogs are administered various doses of aspirin.^{58–64} These studies were not considered relevant to this PICO question because they did not evaluate clinical outcomes. Nonetheless, they address an important consideration regarding the efficacy of aspirin in dogs, because a subset of dogs that received aspirin appeared to be insensitive to its effects.^{59,61,65,66}

No studies directly assessed both the safety of aspirin and its antithrombotic effects; however, dogs were only rarely removed from the included studies due to adverse effects. A single case report (LOE 5, Poor) reported gastrointestinal hemorrhage in a greyhound receiving three 13 mg/kg doses of aspirin.⁶⁷ No experimental studies reported adverse gastrointestinal or renal effects from aspirin, and thus aspirin (even at 100 mg/kg/day for 8 weeks) is considered safe in healthy dogs. Safety in sick dogs has not been evaluated, however, and both comorbidities and concomitant medications are likely to affect its safety profile.

Overall, there is good evidence that aspirin effectively prevents experimental arterial thrombosis in dogs when initiated at least 2 days before a thrombogenic insult such as arterial injury or graft implantation. The reported dose range is very wide, which precludes specific dosing recommendations. It is likely that low and medium oral doses between 0.5 mg/kg/day and 15 mg/kg/day are effective at preventing thromboembolism while maintaining an acceptable safety profile. Although studies have evaluated IV use of aspirin, no commercially available IV formulation is available and thus this is not considered clinically relevant at this time.

Cats

In cats, 14 studies investigated the administration of aspirin. While 7 of these publications were to some degree supportive of the PICO question (LOE 2, 3, and 5, Poor),^{68–74} 6 publications including one randomized clinical trial provided evidence opposing it (LOE 3, Poor, and LOE 1, Good).^{75–80}

A wide range of oral aspirin dosages have been described in cats (from 5 mg/kg twice weekly to 25 mg/kg/day). With these varied dosages, there are varied responses, in outcome studies and following in vitro assessments of platelet function. The FATCAT study (LOE 1, Good) enrolled 75 cats following a cardiogenic thromboembolism and randomized them to receive either oral clopidogrel 18.75 mg once daily or oral aspirin 81 mg every 72 hours.⁷⁵ There was a recurrence of ATE in 75% (27/36) of cats receiving aspirin in the first year compared to 36% (14/39) for those receiving clopidogrel. The median time to a primary event (recurrence of ATE) was 346 days for clopidogrel treated cats compared to 128 days for those receiving aspirin.

A separate retrospective of cats suffering ATE found no significant difference in survival or recurrence rate between cats receiving



high-dose aspirin (≥ 40 mg/cat q72h) and cats receiving low-dose aspirin (5 mg/cat q72h), with recurrence rate of 28% and 20%, respectively (LOE 4, Poor).⁸⁰ In another retrospective study, aspirin was used in 12 of the 17 cats surviving their initial ATE. Four cats went on to suffer a recurrence during the follow up period with 2 in the aspirin group (16% for those receiving aspirin) and 2 not receiving antithrombotic (LOE 4, Poor).⁷⁹

In experimental feline thrombosis models, aspirin at 10 mg/kg IV was able to attenuate markers of platelet activation (thromboxane B₂ (TXB₂), 6-keto prostaglandin F_{1 α} (6k-PGF_{1 α}), prostaglandin E₂ (PGE₂) when collagen was infused intraaortically (LOE 2, Poor).⁶⁸ Aspirin dose adjusted based upon aggregation (typically 12–15 mg/kg PO q24h) was found to be more effective at limiting pulmonary artery thrombi than 97.5 mg twice weekly dosing or placebo (LOE 2, Fair).⁶⁹

While not addressing the PICO question, it is relevant to note that the effect of aspirin on the results of platelet function tests varies considerably in cats. One study (LOE 3, Poor) found 5 mg/kg aspirin PO q48h to have no impact on aggregation as assessed by whole blood impedance aggregometry (10 μ M ADP or 5 μ g/mL collagen), serotonin release, and buccal mucosal bleeding time, although there was a lower serum TXB₂.⁷⁶ Others have documented 81 mg PO q72h inhibits aggregation (whole blood aggregometry) when stimulated with arachidonic acid (AA, 10⁻² M), but not collagen (1 and 3 μ g/mL) or ADP (2 μ M).⁷¹ Mixed results on ADP-induced aggregation were noted when cats were given 5 mg/kg aspirin PO q24h with dietary fatty acids.⁷² Reductions in platelet function were not identified when cats were treated with 5 mg PO q24h or 20.25 mg PO q72h and tested with Plateletworks (ADP, collagen, and AA), multiplate aggregometry, and PFA-100 (collagen/ADP).⁸¹

Adverse gastrointestinal (GI) effects of aspirin were specifically addressed in 2 studies. Only 1 of 5 cats was found to have GI erosions (considered minor) when dosed with 20 mg/kg acetylsalicylate IV and visualized with endoscopy 8 hours later (LOE 3, Poor).⁷⁷ Another study assessed 20 mg/kg of plain aspirin PO versus 100 mg of enteric-coated aspirin PO and found significant gastric, duodenal, and small intestinal lesions after 7 days (LOE 3, Poor).⁷⁸ Dosing 3 hours after an evening meal was best tolerated in this cohort, and cimetidine or misoprostol limited the extent of gastric lesions but not duodenal lesions.⁷⁸

In conclusion, aspirin should not be used as a sole antithrombotic in cats at risk of cardiogenic thromboembolism, with consideration given to clopidogrel in lieu of aspirin in this population. Indeed, aspirin treatment in cats at risk of cardiogenic thromboembolism is associated with a high risk of recurrence, ie, 75% in the first year. There are conflicting studies regarding the impact of aspirin on platelet function ex vivo. This may in part be attributable to varying dosages and dosing intervals, along with differing test methodologies and platelet agonists.

Knowledge gaps

There is inadequate evidence to assess the efficacy of aspirin in dogs in clinical situations predisposing to both ATE and VTE. Similarly, the clinical repercussions of the apparent in vitro lack of sensitivity to aspirin of a subset of dogs are unclear. As pointed out in Domain 2, there is

a need for studies comparing the efficacy of aspirin with clopidogrel in dogs at risk of thrombosis (Guideline 2.5). Further studies are also needed to determine if a more effective dose of aspirin exists for cats while still avoiding adverse effects (eg, gastrointestinal ulceration and risk of hemorrhage), notably if given in addition to clopidogrel. Further studies are also needed to determine the optimal test for assessment of platelet inhibition by aspirin in cats.

PICO QUESTION: Clopidogrel therapy

Guidelines

3.3 Clopidogrel (Dogs)

- We recommend clopidogrel at 1.1–3 mg/kg PO q24h for the prevention of ATE in dogs.
- We suggest a single oral loading dose (eg, 4–10 mg/kg) may be useful for obtaining therapeutic plasma concentrations more rapidly
- No recommendations can be made for, or against, use of clopidogrel as sole agent for VTE in dogs.

3.4 Clopidogrel (Cats)

- We recommend clopidogrel at 18.75 mg PO q24h, for prevention of ATE in cats.
- We suggest a single oral loading dose (eg 37.5 mg total) may be useful for obtaining therapeutic plasma concentrations more rapidly.
- No recommendations can be made for, or against, use of clopidogrel for VTE in cats.

Evidence summary

Dogs

In dogs, 11 papers (LOE 3, Good),^{82–92} and 6 papers (LOE 3, Fair),^{93–98} suggest efficacy of clopidogrel. Of these publications, 10 papers suggest in vivo efficacy against provoked arterial thrombosis, while 7 publications suggest clopidogrel has significant platelet inhibitory effects ex vivo using tests such as platelet aggregometry. The dosages and routes of administration of clopidogrel varied considerably in these publications. The lowest effective dose (effective dose for 50% of the population) was reported to be 0.21 mg/kg PO q24h,⁹² while in other model systems up to 10 mg/kg IV was not consistently efficacious for prevention of arterial thrombosis.⁹¹ The efficacy of a specific dose appears to be dependent on the dog, the model system used and the nature of the stimulus provoking thrombosis or platelet activation.

In vitro tests suggest that 1.1 mg/kg PO q24h causes inhibition of ADP-dependent platelet inhibition in dogs,⁹⁵ while 4 mg/kg PO q24h after a 10 mg/kg loading dose provides protection against arterial thrombosis provoked by bioprosthesis implantation.⁹⁴ In coronary or femoral arterial thrombosis models, doses of 0.5–2.0 mg/kg IV provide protection from thrombosis.^{82,83,90}



The oral bioavailability of clopidogrel in dogs is reported to be 10%.⁹⁹ As such, the oral dosages that are reportedly effective may produce substantially lower plasma concentrations than the intravenous dosages reported elsewhere.

Overall, clopidogrel given orally at a dosage of 1.1–4 mg/kg q24h appears to be safe and may be efficacious for prevention of arterial thrombosis in dogs. An oral loading dose (such as double the maintenance dose or up to 10 mg/kg PO q24h) may be useful for obtaining therapeutic plasma concentrations more rapidly. Similarly, clopidogrel given intravenously at a dosage of 0.5–2.0 mg/kg prior to arterial injury appears to be safe and may be efficacious for prevention of arterial thrombosis in dogs.

Cats

In cats, 8 papers (LOE 1, Good, LOE 2, Fair, and 6 LOE 3–4, Good–Poor) suggest efficacy of clopidogrel.^{75,81,100–105} Of these publications, 3 papers (LOE 1, Good, LOE 3, Good, LOE 4, Poor) suggest in vivo efficacy against provoked arterial thrombosis,^{75,100,101} while 5 publications (LOE 2, Fair, LOE 3, Good, LOE 4, Good) suggest clopidogrel has significant platelet inhibitory effects *ex vivo* using tests such as platelet aggregometry.^{81,102–105} There is some variation in the tests and agonist dosages used to assess the efficacy of clopidogrel in the *in vitro* tests reported in these papers, which may go some way to explaining variations in the efficacy reported.

In contrast to the canine papers, there is very good consistency in the dosages used in the feline studies. The dosage used in most of the feline studies, including the LOE 1 randomized controlled trial (RCT),⁷⁵ was 18.75 mg (total) PO q24h. For a typical cat weighing 3–6 kg, this equates to a dose range of 3.13–6.25 mg/kg. A prior study by Hogan (2004) suggested that 18.75 mg total was not significantly different in efficacy than 37.5 mg or 75 mg PO q24h.¹⁰⁰ One publication suggested that clopidogrel 10 mg PO q24h might not be significantly different in efficacy compared to 18.75 mg PO q24h,¹⁰³ however, this conclusion was based on *ex vivo* platelet function test data. In addition, the authors did recommend a loading dosage. As such, given that 18.75 mg PO q24h appears to be effective and safe, there is currently no rationale to recommend using a lower dosage.

The Hogan et al RCT of clopidogrel versus aspirin in feline ATE was not placebo-controlled, likely for ethical reasons.⁷⁵ However, if one were to assume that aspirin had not beneficial impact at all (and hence behaved like placebo), then clopidogrel was still demonstrably more effective at preventing arterial thrombosis and for prolonging time to thrombosis or cardiac related death. This provides excellent evidence that clopidogrel should be used in cats with ATE, in preference to aspirin. There is virtually no evidence to determine whether aspirin should or should not be added to clopidogrel for ATE management. One retrospective study¹⁰¹ suggested that receiving aspirin and/or clopidogrel was a positive prognostic indicator in cats with ATE, but the data presented was too limited to enable any further exploration of this finding.

Globally, clopidogrel given orally at a dosage of 18.75 mg q24h appears to be safe and efficacious for prevention of arterial thrombosis in cats. An oral loading dose (such as double the maintenance dose)

may be useful for obtaining therapeutic plasma concentrations more rapidly.

Knowledge gaps

While clopidogrel appears safe and effective for the prevention of arterial thrombosis in dogs and cats, no evidence for or against its efficacy in venous thrombosis was identified. Strong evidence supports the use of clopidogrel in cats at risk of cardiogenic ATE, but the canine literature remains insufficient to make a similar claim in dogs at risk of thrombosis and further clinical studies are therefore warranted (eg, IMHA, PLN). As highlighted before, the question remains whether dual antiplatelet therapy would be preferable compared to clopidogrel alone, both in dogs and in cats. As novel antiplatelet agents become available (eg abciximab, ticagrelor, prasugrel), further studies will be needed to establish their safety profile and clinical efficacy.

PICO QUESTION: Warfarin therapy

Guidelines

3.5 Warfarin (Dogs)

- a. We suggest that warfarin should not be used in dogs because it inconsistently improves outcomes and is commonly associated with bleeding complications.

3.6 Warfarin (Cats)

- a. No evidence-based recommendations can be made regarding the use of warfarin in cats at risk for thrombosis.
- b. We suggest that warfarin should not be used in cats because of marked interindividual variation coupled with a narrow therapeutic index.

Evidence summary

Dogs

In dogs, 10 papers (LOE 1, 3, and 5, Poor to Good) suggest efficacy of warfarin.^{106–115} Of these publications, 2 papers document that prophylactic use reduces venous thrombus formation (LOE 3, Fair),¹¹¹ and increases patency of a venous stent (LOE 3, Good)¹⁰⁹ compared to control. Arai et al (LOE 5, Fair)¹¹⁴ and Lantz et al (LOE 3, Fair)¹¹² also suggest some efficacy of warfarin used prophylactically to prevent thrombosis in dogs undergoing bioprosthesis valve replacement and cranial vena cava graft, respectively. However, their conclusions are confounded by the fact that all dogs also received aspirin and a control group was not evaluated. Two papers (both LOE 5, Fair) suggested efficacy of warfarin in the treatment of acute arterial thrombosis,¹¹⁵ and aortic thrombosis,¹¹³ but again the lack of a control group and the coadministration of antiplatelet drugs weakens the conclusions that can be drawn. In an experimental canine model using filter entrapped venous thrombi (partial obstruction of blood flow), Hoffman et al documented that the treatment with warfarin accelerated lysis of mature



venous thrombi compared to no treatment (LOE 3, Good).¹¹⁰ Finally, 3 papers (all LOE 3, Good),^{106–108} support that warfarin at dose ranging from 0.05 to 1 mg/kg q24h is associated with fairly consistent in vitro anticoagulant efficacy described as an international normalized ratio (INR) of 2.0–4.0, depending on the study.

However, 7 papers document a lack of efficacy of warfarin in preventing thrombosis or bleeding complications (LOE 3.5; Poor-Good).^{32,106,111,112,116–118} Of these publications, 4 studies suggest that warfarin lacks efficacy to prevent thrombosis of mitral valve prosthesis,^{117,118} to maintain patency of venous graft compared to no treatment,¹¹⁶ or patency of an intracoronary arterial stent compared to aspirin/dipyridole.³² Four studies also document hemorrhagic complications, often deadly, with relatively common warfarin dosage of 0.17–1.5 mg/kg q24h,^{32,111,112,116} although no prothrombin time (PT) monitoring was performed in 2 of these studies.

Several studies did not address the above PICO question, but did highlight the narrow therapeutic index of warfarin. In dogs receiving long-term warfarin therapy, changes in the pharmacology of the drug over time has been documented.¹¹⁹ In other words, clinical anticoagulation may require multiple dosage adjustments despite the early attainment of apparently therapeutic anticoagulation on a fixed dosage schedule. Numerous drugs, including glutethimide, amobarbital, secobarbital, meprobamate, amiodarone, halofenate, chlordan, phenobarbitone, phenylbutazone, and vitamin E, have been documented to interact with warfarin.^{107,120–126} Of great clinical significance, Shen et al documented that the coadministration of aspirin increases the concentration of warfarin and associated prothrombin time, which may therefore increase the risk of bleeding.¹⁰⁶ Finally, warfarin is 93% protein-bound in dogs,¹²⁷ which may need to be considered when managing individual patients.

Owing to the narrow therapeutic index of warfarin, close monitoring of its anticoagulant effect using PT is emphasized throughout the literature. However, the appropriate loading dose and therapeutic targets necessary to avoid bleeding complications while producing clinically adequate anticoagulation have yet to be determined. In papers supporting the use of warfarin prophylactically or therapeutically for thrombi dissolution, and without adverse reactions (Makutani 1999 and Winter 2012, respectively),^{109,113} the starting dose varied from 0.05 to 0.5 mg/kg q24h with either an antithrombotic target to prolong PT by 1.5–2-fold normal,¹⁰⁹ or to attain an INR of 2.0–3.0.¹¹³ Similarly, loading doses ranging from 0.05 mg/kg to 0.3 mg/kg have been associated with in vitro anticoagulant efficacy defined as an INR 2.0–3.0, an INR 2.0–4.0, or 1.5–2 times baseline PT, without bleeding complications.^{107,108,127–129} A pharmacokinetic study in healthy dogs also suggested that the dosage interval should be reduced 12 hours,¹²⁷ but this recommendation has not been applied by most subsequent studies. The most explicit dosage regime is provided by Monnet and Morgan, who evaluated the effect of 3 loading doses of warfarin on the INR in dogs in a model of postbilateral iliac artery grafting.¹⁰⁸ They concluded that a warfarin loading dose of 0.2 mg/kg for 2 days should induce an increase of the INR to a value between 2.0 and 3.0, during which period heparin should be administered concurrently. After 2 days, they recommended discontinuation of heparin, and a reduction in the loading dose of warfarin by half. Thereafter, the patient should be

monitored daily by evaluation of the INR, aiming for a range of 2.0–3.0. In that study, no bleeding tendencies were noted, but the patency of the iliac artery grafts was not discussed. It should be noted that using a similar warfarin loading dose of 0.2–0.3 mg/kg for 3–5 days, Wheat et al, reported wide variation in the resulting INR in 9 dogs, ranging from an INR <2 in 2 dogs to >3.5 in 4 dogs, although without bleeding complications.¹²⁹

Interestingly, in studies suggesting that warfarin is either ineffective for prevention of thrombosis or is associated with bleeding complications, the starting dose tended to be higher (1–1.5 mg/kg q24h),¹¹¹ not clearly defined,^{116,117} or described without clear protocolized monitoring of PT.^{32,112,118}

Overall, in dogs at risk of arterial or venous thrombosis, the use of warfarin to improve outcome is inconsistent and bleeding complications are commonly reported, especially if close monitoring is lacking. If warfarin is to be used prophylactically or therapeutically to dissolve thrombosis, a conservative loading dose is imperative (eg, 0.05–0.2 mg/kg q24h) as well as close initial and ongoing monitoring as described in Domain 4 (Guideline 4.2).

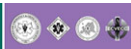
Cats

In cats, publications are scarce concerning the use of warfarin, and only 2 papers suggest some degree of efficacy.^{130,131} Pouchelon et al, (LOE 5, Poor) reported the case of a cat diagnosed with pulmonary thromboemboli and treated with warfarin for a total of 20 days, which led to its clinical improvement without adverse reaction.¹³¹ On day 5, the starting dose of 0.086 mg/kg PO was reduced to every other day administration, because PT was prolonged 7 times baseline. The lack of control and coadministration of heparin during the course of treatment limits the conclusions that can be drawn, however.¹³¹ An experimental pharmacodynamic study in 10 healthy cats using a single IV dose of 0.5 mg/kg of racemic warfarin demonstrated that warfarin is able to block prothrombin complex synthesis as demonstrated by significant prolongation of PT.¹³⁰ A predicted median daily dose to yield therapeutic concentrations of 0.06–0.09 mg/kg per day was recommended (based on INR of 2.0–3.0). However, marked interindividual variation and a narrow therapeutic index were reported, highlighting the need for close monitoring.

Other studies available either opposed the use of warfarin in cats,^{132,133} (LOE 3, Good-Fair) or do not reveal any benefit,¹³⁴ (LOE 5, Poor). The administration of warfarin at a dosage sufficient to prolong PT by 2–4 times baseline, alone or in combination with aspirin, had no protective effect in an experimental model of arterial thrombosis in cats.¹³² Similarly, warfarin administered at a dose of 0.05 mg/kg PO q12h for 9 days in a cat diagnosed with aortic thromboembolism did not appear to influence outcome.¹³⁰ Of clinical interest, it was demonstrated that the warfarin is unequally distributed within a commercial tablet that may increase the risk of over or under dosing our feline patients (or small dogs).¹³³

Knowledge gaps

While in dogs there is sufficient evidence against the use of warfarin (lack of efficacy and/or high complication rates), there is a scarce body



of evidence regarding the use of warfarin compared to no therapy or other antithrombotic therapies in cats with a risk of thrombosis. In both species, narrow therapeutic indices have been documented, highlighting the need for close clinical monitoring and repeated PT assessment. Unfortunately, no data are available to predict an appropriate target range.

PICO QUESTION: Unfractionated Heparin therapy

Guidelines

3.7 Unfractionated Heparin (Dogs)

- a. UFH can be effectively administered by the IV or SC routes in dogs.
- b. Optimal UFH dose likely varies in individual dogs to maximize antithrombotic effects and minimize hemorrhagic complications.
- c. We suggest an initial IV dosing scheme of 100 U/kg bolus, then 480–900 U/kg/24h (20–37.5 U/kg/h) constant rate infusion in dogs.
- d. We suggest an initial SC dosage of UFH of 150–300 U/kg q6h in dogs.
- e. We recommend that UFH is not administered by inhalation or PO in dogs.

3.8 Unfractionated Heparin (Cats)

- a. Only a SC route of administration of UFH has been investigated in cats.
- b. We suggest an initial SC dosage of UFH of 250 U/kg q6h in cats.

Evidence summary

Dogs

The apparent antithrombotic efficacy of unfractionated heparin (UFH) is substantiated by 8 publications (LOE 1–4, Fair-Good).^{128,135–141} While UFH administered via several routes appears potentially effective, the optimal dosing scheme is unestablished based on the current literature base. Subcutaneous^{135,140,142} (LOE 1 and 3, Good and Fair, respectively) and intravenous¹⁴¹ (LOE 4, Fair) routes are both acceptable. Oral administration of UFH was shown to induced changes in coagulation diagnostics (eg, anti-Xa activity, aPTT (activated partial thromboplastin time), and antithrombin), but its biological antithrombotic effects have not been documented to date (LOE 3, Fair).¹⁴³ Inhaled unfractionated heparin does not appear effective in dogs however (LOE 3, Fair).¹⁴⁴ For intravenous administration, a bolus dose of unfractionated heparin (100 U/kg) followed by constant rate infusion (480–900 U/kg/day) is reasonable, with escalation or deescalation of this dose depending on the results of therapeutic monitoring.^{141,145,146}

A starting subcutaneous dose in the range of 150–300 U/kg q6h, with subsequent dose adjustment based on monitoring, is recommended.^{135,140,142,147} The optimal dose of unfractionated heparin likely varies in individual patients and dose adjustment based on appropriate coagulation monitoring tests appears effective in dogs (LOE 1, Good).¹³⁵ Similarly, certain patient populations (eg, IMHA) may benefit from considerably higher doses than these starting points (LOE 1, Good; LOE 5, Fair).^{135,147} Therapeutic monitoring is recommended irrespective of the route of administration. In that regard, several publications gauge apparent dosing efficacy by the ability to achieve the target unfractionated heparin specific anti-Xa activity range (0.35–0.7 U/mL) (Guideline 4.3b).

There is a scarce body of evidence regarding the use of unfractionated heparin in cats, and none in cats at risk of thrombosis, with only one study (LOE 3, Fair),¹⁴⁸ investigating the apparent antithrombotic effect of unfractionated heparin compared to low molecular weight heparin in healthy cats. Relatively minor hemorrhage may occur with unfractionated heparin irrespective of the dosing scheme, reiterating another advantage of therapeutic monitoring in small animals. A reasonable starting subcutaneous dose for unfractionated heparin in cats is 250 U/kg q6h.¹⁴⁸

Cats

With only one study available in healthy cats, further studies are warranted to establish the most appropriate starting dose of UFH in a feline population at risk of thrombosis and to investigate alternative routes of administration. That said, a consistent fixed dose of unfractionated heparin for dogs and cats likely does not exist. Individual dose adjustment, based on therapeutic monitoring, is advised and may be associated with improved clinical outcome. In that regard, a knowledge gap exists regarding the importance of anti-Xa activity achievement with respect to clinical efficacy in small animals (Guideline 4.3bc). There are insufficient data to support a clear superiority or inferiority of either UFH over low molecular weight heparin in dogs, and even less so in cats (Guidelines 2.9 and 2.10).

Knowledge gaps

With only one study available in healthy cats, further studies are warranted to establish the most appropriate starting dose of UFH in a feline population at risk of thrombosis and to investigate alternative routes of administration. That said, a consistent fixed dose of unfractionated heparin for dogs and cats likely does not exist. Individual dose adjustment, based on therapeutic monitoring, is advised and may be associated with improved clinical outcome. In that regard, a knowledge gap exists regarding the importance of anti-Xa activity achievement with respect to clinical efficacy in small animals (Guideline 4.3bc). There are insufficient data to support a clear superiority or inferiority of either UFH over low molecular weight heparin in dogs, and even less so in cats (Guidelines 2.9 and 2.10).

PICO QUESTION: Dalteparin therapy

Guidelines

3.9 Dalteparin (Dogs)

- a. We suggest an initial SC dosage of 100–175 U/kg q8h in dogs.
- b. Minor bleeding may be noted at the doses reported above, but serious bleeding is unlikely.

3.10 Dalteparin (Cats)

- a. In cats, frequent SC administration is likely necessary for maintenance of the human target anti-Xa activity range.
- b. We suggest lower dosages compared to dogs may be acceptable at increased frequency eg, 75 U/kg SC q6h.
- c. Bleeding complications, usually minor and self-limiting, may occur with a variety of dosing schemes.



Evidence summary

Dogs

Despite several publications, little data exist regarding the influence of dosing scheme on clinical efficacy or complications of dalteparin in small animals.^{142,146,149–153} Only 4 studies suggest some degree of efficacy of dalteparin in dogs (LOE 3, 4, and 5, Poor and Fair),^{142,150,151,153} as other studies were considered neutral to the PICO question.^{146,149,154} It is to be noted that several manuscripts share the common assumption that the human target therapeutic anti-Xa activity range (eg, 0.5–1.0) is an appropriate target for dogs,^{141,149–151,154,155} and may correlate to clinical antithrombotic effect. Bearing in mind this concept, doses of 100–175 U/kg SC may result in achievement of the human anti-Xa activity range,^{149–151,154,155} with consistent achievement more likely if administration occurs three times daily.¹⁴⁹ Bleeding complications in dogs appear uncommon, although clear overdosage of dalteparin may result in severe potentially life-threatening bleeding (LOE 5 poor),¹⁵¹ while most bleeding events otherwise appear relatively minor (eg, hematuria, venipuncture site bleeding).¹⁵¹ Dalteparin may be associated with fewer bleeding complications than high-dose unfractionated heparin (ie, bolus of 100 U/kg of UFH IV, then constant rate infusion of 900 U/kg/day), although the antithrombotic effect of dalteparin at this dose (ie, 100 U/kg SQ q12h) is uncertain (LOE 4 fair).¹⁴¹

Cats

In cats, 6 publications address the usage of dalteparin, but were considered neutral to the PICO question listed above (LOE 3 and 5, Poor and Fair).^{74,148,152,156–158} Frequent dosing may be appropriate, yet this is typically derived from studies performed in a small number of cats or based on pharmacokinetic models.^{148,152,156} Based on the limited literature, a dosing scheme of 75–150 U/kg administered subcutaneously q6h appears to be a reasonable dosing strategy (LOE 3, Poor, and Fair).^{148,156} Minor bleeding may be noted even at the clinical doses reported above, but serious bleeding is unlikely unless severe overdosing occurs.^{148,156,157}

Knowledge gaps

There is incomplete information in the current literature to confidently answer the above PICO questions concerning the use of dalteparin in both dogs and cats. Further studies, ideally RCTs, are warranted to investigate the efficacy and safety of dalteparin in canine and feline patients at risk of thrombosis. A knowledge gap also exists regarding the importance of anti-Xa activity achievement with respect to clinical efficacy in small animals.

PICO QUESTION: Enoxaparin therapy

Guidelines

3.11 Enoxaparin (Dogs)

- We suggest enoxaparin at a dosage of 0.8 mg/kg SC q6h is safe and well tolerated in dogs.
- This dose may not achieve anti-Xa levels considered to be therapeutic in people in all breeds of dog.
- Only minor bleeding complications have been reported in association with enoxaparin use in dogs.

3.12 Enoxaparin (Cats)

- We suggest enoxaparin at a dosage of 0.75–1 mg/kg SC q6–12h should be considered in cats with a risk of VTE.
- We suggest enoxaparin be administered q6h to reduce inter-individual variation in peak anti-Xa activity.

Evidence summary

Dogs

In dogs, 5 papers suggested efficacy of enoxaparin (LOE 3, 4, and 5, Poor to Good),^{142,159–162} and the most commonly used protocol is 0.8 mg/kg SC q6h.^{159–161} Although there is a scarce body of evidence regarding the use of enoxaparin compared to no antithrombotic therapy or other antithrombotic therapies in dogs a risk of thrombosis, there is some evidence suggesting the preferable use of a specific enoxaparin protocol in certain diseases with a risk of thrombosis. For example, the use of enoxaparin at a dose of 0.8 mg/kg SC q6h appeared well tolerated and relatively safe in dogs with presumed primary IHMA.¹⁶⁰ However, this protocol may not be adequate across all breeds, as demonstrated by a lack of efficacy (eg, targeted anti-Xa activity) in healthy adult beagles.¹⁶³

Cats

Publications are scarce concerning the use of enoxaparin in cats, and only 3 papers suggest some degree of efficacy in healthy cats (LOE 3, Fair).^{148,164,165} The use of enoxaparin at 1 mg/kg SC q12h in healthy cats failed to induce sustained anti-Xa activity.¹⁴⁸ However, Van De Wiele et al concluded that enoxaparin, administered at the same dosage (ie, 1 mg/kg SC q12h) in a venous stasis model in healthy cats, produces an antithrombotic effect, but that anti-Xa activity is a poor predictor of that effect.¹⁶⁵ Finally, based on pharmacokinetic data, Mischke et al concluded that a dosage schedule of 0.75 mg/kg four times a day seems suitable for therapeutic use of enoxaparin in cats as it leads to reproducible peak anti-Xa activities within the target range for the treatment of thrombosis in humans with low interindividual variation.¹⁶⁴

Knowledge gaps

There is a scarce body of evidence regarding the use of enoxaparin compared to no antithrombotic therapy or other antithrombotic therapies in dogs or cats with a risk of thrombosis. RCTs, including appropriate monitoring of drug activity, are warranted to establish clear recommendations regarding appropriate protocol and valid comparisons of efficacy in clinical patients.

PICO QUESTION: Fondaparinux therapy

Guidelines

3.13 Fondaparinux (Dogs and Cats)

- No studies of fondaparinux in dogs were identified.
- A dose of fondaparinux of 0.06 or 0.20 mg/kg SC q12h was sufficient to achieve a peak plasma anti-Xa activity in cats considered effective in people, without bleeding complications.

Evidence summary

Dogs and cats

To date, scientific evidence to answer the PICO question is lacking. There are no studies evaluating fondaparinux in dogs and only one study in cats (LOE 3, Fair).¹⁶⁶ The study was a dose-determination study in 6 healthy purpose bred cats targeting anti-factor Xa levels comparative to what is considered effective in humans. Briefly, based upon evaluating anti-Xa activity, the investigators' data revealed that fondaparinux dosage of 0.06 or 0.20 mg/kg SC q12h was sufficient to achieve a peak plasma anti-factor Xa activity in cats that has been deemed therapeutic in people. No bleeding complications were reported.

Knowledge gaps

Basic pharmacokinetics and pharmacodynamics studies in healthy dogs are needed to establish reasonable dosing scheme for fondaparinux. Thereafter, clinical trials evaluating fondaparinux in dogs and cats are warranted.

PICO QUESTION: Rivaroxaban therapy

Guidelines

3.14 Rivaroxaban (Dogs)

- Based on preliminary data, rivaroxaban appears safe and well tolerated in dogs.
- We suggest a dosage of 1–2 mg/kg/day in dogs.

3.15 Rivaroxaban (Cats)

- Based on preliminary data, rivaroxaban appears safe and well tolerated in cats.
- We suggest a dosage of 0.5–1 mg/kg/day in cats.

Evidence summary

Dogs

A total of 3 studies suggest the efficacy of rivaroxaban in dogs (LOE 2 and 4, Fair and Poor),^{167–169} with only 1 study assessing its efficacy in the prevention of thrombosis.¹⁶⁷ Overall, rivaroxaban appears to be safe and well tolerated based upon the available evidence for dogs.

Briefly, Conversy et al assessed the short-term safety and anticoagulant effect of rivaroxaban in healthy dogs (LOE 2, Fair). Twice daily (q 8 hour) dosing of ~2 mg/kg resulted in an anticoagulant effect lasting almost 24 hours (returned to baseline after 17.5–26.8 hours versus 7.9–18.7 hours for once daily administration) as assessed by anti-Xa activity and thrombin generation.¹⁶⁸ The peak anticoagulant effect was seen at 1.5–2 hours, and there was no further increase in peak anticoagulation effect with twice daily administration versus once daily. No adverse effects were noted. Morassi et al (LOE 2, Fair) assessed the safety and tolerability of rivaroxaban in a prospective, positive-controlled, unblinded clinical trial including 24 dogs with primary immune-mediated hemolytic anemia.¹⁶⁷ Patients were scheduled to receive either 0.5–1 mg/kg rivaroxaban PO q24h or dual antiplatelet therapy (2–3 mg/kg clopidogrel PO q24h and 1 mg/kg aspirin PO q 24 h). There was no identified adverse drug reaction, evidence of hemorrhage, significant prolongation in PT/aPTT, increase in transfusion requirement, or difference in rate of thrombotic events, or survival to discharge at 1 month and 3 months. Three out of 5 dogs that did not survive the study period had thrombotic complications, including one dog receiving rivaroxaban (mesenteric thrombosis).

Lastly, therapeutic efficacy of rivaroxaban (0.6–1.25 mg/kg PO q24h) is supported by a single case series of 4 dogs suffering from various thromboses (arterial and venous), which showed an apparent benefit (for thrombus resolution) when switched from other antithrombotics.¹⁶⁹

Cats

Only a pharmacokinetic and pharmacodynamics study is available in healthy cats to base recommendation in that species (LOE 2, Fair).¹⁷⁰ Healthy cats administered single doses of 2.5 and 5 mg of rivaroxaban PO achieved anti-Xa activities of 2.8 ± 1.3 and 3.5 ± 0.6 IU/mL, respectively, with a quick onset (3 hours) and a dose-dependent anticoagulation effect. Anti-Xa activities were maintained near the therapeutic target for thromboprophylaxis in humans at 24 hours (0.3 ± 0.2 and 0.47 ± 0.2 IU/mL, respectively; target of 0.5–1 IU/mL). However, cats achieved target plasma rivaroxaban concentration for thromboprophylaxis in people (140–240 ng/mL) when the anti-Xa activity was 2.6–4.1 IU/mL. There were no adverse effects noted with rivaroxaban administration in the healthy cats.¹⁷⁰

Knowledge gaps

Rivaroxaban appears safe and well tolerated in both dogs and cats, however studies designed to assess the efficacy of rivaroxaban are lacking in animal at risk of thrombosis. In addition, the optimal dose and dosing interval for thromboprophylaxis and the treatment of thrombotic issues requires further study.

CONCLUSION

There is an obvious need for further studies relating to dosing of antithrombotic medications in veterinary medicine. Most notably

randomized controlled clinical trials will be essential to enable formulation of robust evidence-based recommendations concerning specific antithrombotic protocols in companion animals and in specific disease states. Additional studies are particularly warranted for drugs where dosage schemes are currently suggested based on limited literature and the clinical expertise of the panel.

In continuation of aspirin, clopidogrel is now used widely in companion animals, with solid support from the veterinary literature particularly in cats. Novel antiplatelet agents, including novel P2Y₁₂ inhibitors (prasugrel, ticagrelor, and cangrelor)¹⁷¹ and the GPIIb/IIIa inhibitors (abciximab, tirofiban, and eptifibatide),¹⁷² may eventually be included in future iterations of these guidelines as these medications become more affordable. Progress is occurring in the use of anticoagulant therapies also. As a result of the narrow therapeutic index, potential for side effects and drug interactions, the field is moving away from warfarin toward the direct oral anticoagulants. Rivaroxaban is an exciting new drug that appears safe and effective in dogs and cats, and will likely be the subject of further studies in the near future. More work is urgently required in this area to establish the most appropriate protocols for our patients and to confirm the efficacy and safety of the direct oral anticoagulants in clinical practice. Likewise, apixaban,¹⁷³ another novel direct oral Xa-inhibitor, and dabigatran a direct oral thrombin inhibitor,¹⁷⁴ warrant further investigation in small animals.

The phenomenon of individual drug response variability is well-recognized in human medicine,¹⁷⁵ but needs to be better defined in small animals. Individual variability in response to antithrombotics is likely due to various factors including disease state, age, breed, sex and neuter status, nutrition, concomitant medications, and genetics. It is likely the underlying causes of individual drug response variability will not be straightforward to confirm, but this area of veterinary pharmacology demands further attention. Moving forward, new antithrombotic drugs will continue to be investigated in human medicine to optimize their pharmacological properties and improve their benefit-risk profiles. Veterinarians need to do the same before incorporating these drugs into our clinical practice. Overall, significant knowledge gaps were highlighted by this work, which will hopefully drive novel research.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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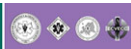
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